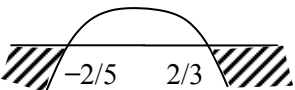
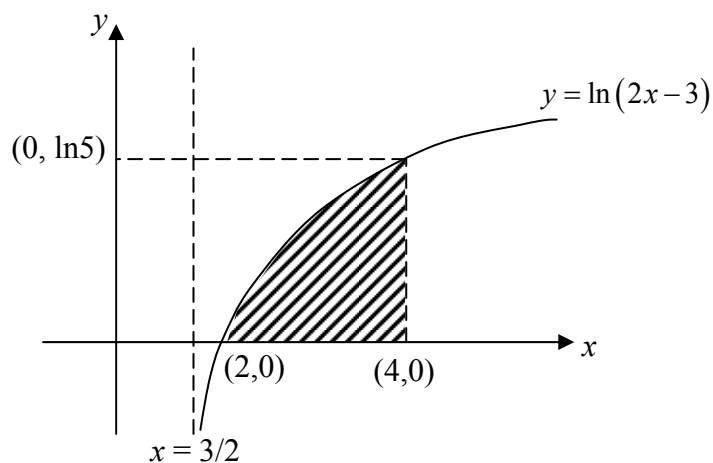
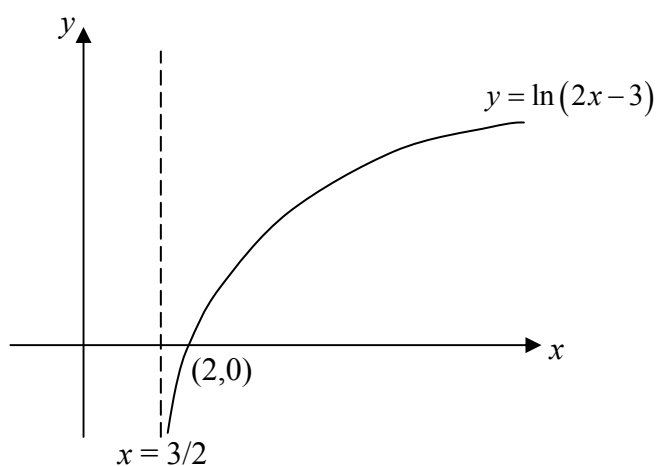


2012 Year 6 H1 Mathematics Preliminary Examination Solutions

	Solutions
1	For $kx^2 + kx + 2x + 4k > 0 \Rightarrow kx^2 + (k+2)x + 4k > 0$ $(k+2)^2 - 4(k)(4k) < 0$ and coefficient of $x^2 = k > 0$ $(k+2)^2 - (4k)^2 < 0$ $(k+2-4k)(k+2+4k) < 0$ $(2-3k)(2+5k) < 0$  $-\frac{2}{5} < k < \frac{2}{3}$ and $k > 0$ The solution set is $\left\{ k \in \mathbb{R} : 0 < k < \frac{2}{3} \right\}$. <u>Alternative</u> $(2-3k)(2+5k) < 0$ $(3k-2)(2+5k) > 0$ 
2	Let length of side AC be y units and the length of side BC be x units. Given that Perimeter of $\triangle ABC = 36$ $x + (x+2) + y = 36$ $2x + y = 34$ $y = 34 - 2x \dots (1)$ Substitute (1) into (2): $2x^2 + 4x + 4 = (34 - 2x)^2$ $2x^2 + 4x + 4 = 1156 - 136x + 4x^2$ $2x^2 - 140x + 1152 = 0$ $x^2 - 70x + 576 = 0$ By GC, $x = 60.475\dots$ (reject, since $x < 36$) or $x = 9.5245\dots$ For $x = 9.5245\dots$, $y = 34 - 2(9.5245) = 14.951\dots$ The length of side AC is 15.0 units (3sf).

Solutions

3



$$y = \ln(2x - 3)$$

$$e^y = 2x - 3$$

$$x = \frac{e^y + 3}{2}$$

Area of shaded region

$$= 4 \ln 5 - \int_0^{\ln 5} \left(\frac{e^y + 3}{2} \right) dy$$

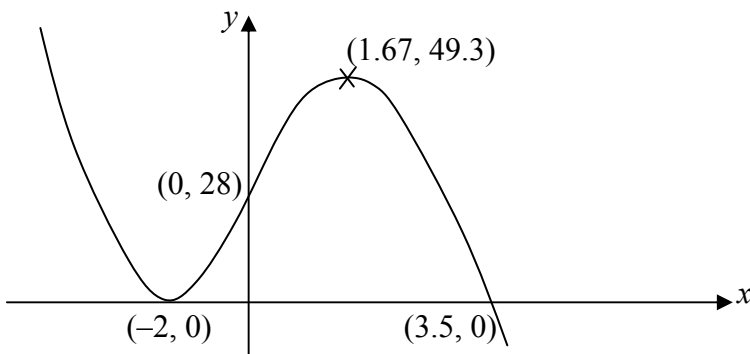
$$= 4 \ln 5 - \frac{1}{2} \int_0^{\ln 5} (e^y + 3) dy$$

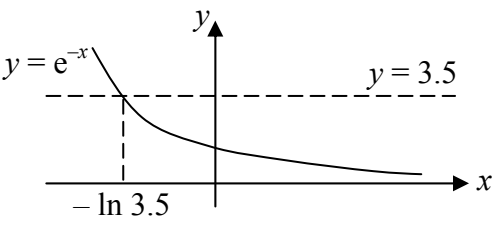
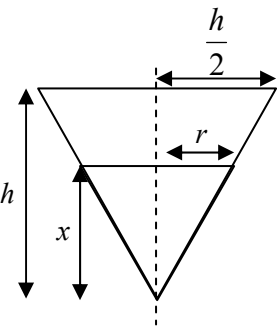
$$= 4 \ln 5 - \frac{1}{2} [e^y + 3y]_0^{\ln 5}$$

$$= 4 \ln 5 - \frac{1}{2} [(e^{\ln 5} + 3 \ln 5) - e^0]$$

$$= 4 \ln 5 - \frac{1}{2} (5 + 3 \ln 5 - 1)$$

$$= \left(\frac{5}{2} \ln 5 - 2 \right) \text{ units}^2$$

	Solutions
4(i)	$f'(x) = \frac{2}{bx+3}$ $f'(2) = \frac{2}{2b+3} = 2$ $4b+6=2$ $b=-1$ $\therefore f'(x) = \frac{2}{3-x}$ $f(x) = \int \left(\frac{2}{3-x} \right) dx = -2 \ln(3-x) + C$ Given that $f(-1) = 0$, $-2 \ln(4) + C = 0$ $C = 2 \ln 4$ $\therefore y = -2 \ln(3-x) + 2 \ln 4$
4(ii)	$f(x) = -2 \ln(3-x) + 2 \ln 4$ For $f(x+2) = f(x) + f(2)$, $-2 \ln(3-(x+2)) + 2 \ln 4 = [-2 \ln(3-x) + 2 \ln 4] + [-2 \ln(3-2) + 2 \ln 4]$ $-2 \ln(1-x) = -2 \ln(3-x) + 2 \ln 4$ $\ln(3-x) - \ln(1-x) = \ln 4$ $\ln \left(\frac{3-x}{1-x} \right) = \ln 4$ $\frac{3-x}{1-x} = 4$ $3-x = 4-4x$ $3x = 1$ $x = \frac{1}{3}$
5(i)	$y = 28 + 20x - x^2 - 2x^3$ 
5(ii)	The range of values of k is $k < 0$ or $k > 49.3$.

	Solutions
5(iii)	For $y = \ln(28 + 20x - x^2 - 2x^3)$ to exist, $28 + 20x - x^2 - 2x^3 > 0$. $\therefore$ From the graph, $x < -2$ or $-2 < x < 3.5$
	For $y = \ln\left(28 + \frac{20}{e^x} - \frac{1}{e^{2x}} - \frac{2}{e^{3x}}\right)$ to exist, $28 + \frac{20}{e^x} - \frac{1}{e^{2x}} - \frac{2}{e^{3x}} > 0$. Replace x by e^{-x}, $e^{-x} < -2$ or $-2 < e^{-x} < 3.5$ (no solution $0 < e^{-x} < 3.5$) $\therefore e^{-x} > 0$ for all $x \in \mathbb{R}$)  From the graph, $x > -\ln 3.5$
6(ai)	Let the height of the cone be h cm. Using similar triangles, $\frac{x}{h} = \frac{r}{h/2} \Rightarrow r = \frac{x}{2}$ $V = \frac{1}{3}\pi\left(\frac{x}{2}\right)^2 x$ $= \frac{1}{12}\pi x^3 \text{ (shown)}$ 
6(aii)	$\frac{dV}{dx} = \frac{1}{4}\pi x^2$ and $\frac{dV}{dt} = -3$ $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$ When $x = 6$, $\frac{dV}{dx} = \frac{1}{4}\pi(6^2) = 9\pi$ $-3 = 9\pi \frac{dx}{dt}$ $\frac{dx}{dt} = -\frac{1}{3\pi}$ Hence, rate of decrease of depth of water is $\frac{1}{3\pi} \text{ cm s}^{-1}$.
6(bi)	$h = -t^4 + \frac{28}{3}t^3 - 30t^2 + 36t + 1$ $\frac{dh}{dt} = -4t^3 + 28t^2 - 60t + 36$ For stationary points, $\frac{dh}{dt} = 0$ $-4t^3 + 28t^2 - 60t + 36 = 0$

Solutions

From GC, $t = 1$ or $t = 3$.

For $t = 1$,

	1^-	1	1^+
$\frac{dh}{dt}$	>0	0	<0
Slope of tangent	/	—	\

$\therefore$ At $t = 1$, turning point is a maximum point.

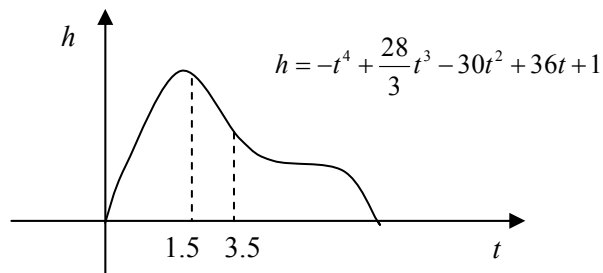
For $t = 3$,

	3^-	3	3^+
$\frac{dh}{dt}$	<0	0	<0
Slope of tangent	\	—	\

$\therefore$ At $t = 3$, turning point is a stationary point of inflexion.

Required values: $t = 1, h = \frac{46}{3}$.

6(bii)



Greatest height is attained at $t = 1.5$ (since h is decreasing within the time interval).

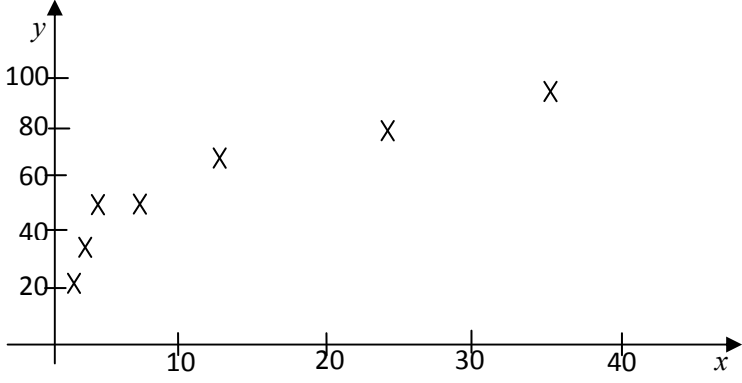
Greatest height attained

$$= -(1.5)^4 + \frac{28}{3}(1.5)^3 - 30(1.5)^2 + 36(1.5) + 1$$

$$= 13.9375\text{m}$$

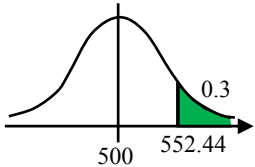
	Solutions		
7	Given $X \sim N(\mu, \sigma^2)$ $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{50}\right)$ and $X_1 + \dots + X_{50} \sim N(50\mu, 50\sigma^2)$ Given $P(\bar{X} < 128) = 0.15$ and $P(X_1 + \dots + X_{50} > 7500) = 0.25$ Standardizing, $Z \sim N(0,1)$ <table border="0"> <tr> <td style="vertical-align: top;"> $P\left(Z < \frac{128 - \mu}{\sigma/\sqrt{50}}\right) = 0.15$ $\frac{128 - \mu}{\sigma/\sqrt{50}} = -1.03643$ $\mu\sqrt{50} - 1.03643\sigma = 128\sqrt{50} \dots(1)$ </td><td style="vertical-align: top;"> $P\left(Z > \frac{7500 - 50\mu}{\sqrt{50}\sigma}\right) = 0.25$ $\frac{7500 - 50\mu}{\sqrt{50}\sigma} = 0.674489$ $50\mu + 0.674489\sqrt{50}\sigma = 7500 \dots(2)$ </td></tr> </table> Solving equation (1) and (2), using GC, $\mu = 141.327 \dots = 141$ (nearest whole number) $\sigma = 90.923 \dots = 91$ (nearest whole number)	$P\left(Z < \frac{128 - \mu}{\sigma/\sqrt{50}}\right) = 0.15$ $\frac{128 - \mu}{\sigma/\sqrt{50}} = -1.03643$ $\mu\sqrt{50} - 1.03643\sigma = 128\sqrt{50} \dots(1)$	$P\left(Z > \frac{7500 - 50\mu}{\sqrt{50}\sigma}\right) = 0.25$ $\frac{7500 - 50\mu}{\sqrt{50}\sigma} = 0.674489$ $50\mu + 0.674489\sqrt{50}\sigma = 7500 \dots(2)$
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8(a)	Sampling interval, $k = \frac{810}{30} = 27$ List the <u>applications in order of the applicants' names</u> (or loan amount), and <u>randomly select the first application</u> to process. Subsequently, <u>select every 27th application</u> until a sample of 30 applications is selected, <u>going back to the front of the list if necessary</u>.		
7(b)	A systematic random sample may not ensure that all categories of loan applications were processed within that day if the <u>sampling interval coincides with a cyclic pattern</u> in the list of applications.		
7(c)	Stratified random sampling ensures that each category of loan is <u>proportionately represented</u> in the sample.		

	Solutions
9(i)	Unbiased estimate of population mean is $\bar{t} = \frac{123}{120} + 15 = 16.025$
9(ii)	The term “unbiased estimate” means that the expected value of the sample statistic is equal to the population parameter that is being measured.
9(iii)	Test $H_0 : \mu = 15$ Against $H_1 : \mu \neq 15$ Conduct a 2-tailed test at 5% significance level. Since $n = 120$ is large, by Central Limit Theorem and under H_0, $\bar{T} \sim N\left(15, \frac{4.47^2}{120}\right) \text{ approximately.}$ Since $\bar{t} = 16.025$, $p\text{-value} = 0.0120$ (3 s.f) Since $p\text{-value} = 0.0120 < 0.05$, we reject H_0 and conclude that there is sufficient evidence at 5% significance level that the mean waiting time differs from 15 minutes.
9(iv)	Since n is large, by the Central Limit Theorem, $\bar{T} \sim N\left(15, \frac{4.47^2}{n}\right) \text{ approximately}$ Since H_0 is rejected, $p\text{-value} < \alpha$ $2P(\bar{T} > 15.5) < 0.06$ $P(\bar{T} > 15.5) < 0.03$ Standardizing, $Z \sim N(0,1)$ $P\left(Z > \frac{15.5 - 15}{\frac{4.47}{\sqrt{n}}}\right) < 0.03$ $\frac{0.5\sqrt{n}}{4.47} > 1.8807...$ $n > 282.69...$ $n \geq 283, \text{ where } n \in \mathbb{Z}^+.$ Least value of $n = 283$.

Solutions	
10(i)	 From GC, $r = 0.92152... = 0.922$ (3 s.f)
10 (ia)	From GC, regression line of y on x is $y = 1.8308x + 35.030$ $y = 1.83x + 35.0$ (3 s.f)
10 (ib)	From GC, regression line of x on y is $x = 0.46384y - 14.438$ $x = 0.464y - 14.4$ (3 s.f)
10(iii)	The gradient of the regression line of y on x represents the estimated increase in test score per hour spent on studying.
10(iv)	Using the regression line of y on x, $60 = 1.8308x + 35.030$ Estimated value of $x = 13.638...$ $= 13.6$ (3 s.f) The regression line of x on y is not suitable as the test score depends on the number of hours spent studying.
10(v)	Let the mean number of hours spent studying and mean score of the 53 students be p and q respectively. $y = 1.6x + 35$ $x = 0.5y - 14 \Rightarrow y = 2(x + 14)$ Solving for point of intersection from GC, $(p, q) = (17.5, 63)$ Mean number of hours studied $= \frac{1 + 2 + 3 + 6 + 12 + 24 + 36 + 53(17.5)}{60}$ $= 16.858...$ $= 16.9$ (3 s.f) Mean score of 60 students $= \frac{20 + 33 + 49 + 52 + 70 + 80 + 95 + 53(63)}{60}$ $= 62.3$

Qn	Suggested Solution									
11(i)	$P(X = 0) + P(X = 19) = 0.1$ $(1 - p)^{20} + \binom{20}{19} p^{19} (1 - p) = 0.1$ By GC, $p = 0.10874 \dots = 0.109$ (3 s.f.)									
11(ii)	Reasons should address either one of these behaviour: 1: <i>The probability of "success" p is the same for each outcome.</i> 2: <i>The responses are independent of each other.</i> <table><tr><th></th><th>Behaviour of X</th><th>Behaviour of Y</th></tr><tr><td>1.</td><td>Probability of responding positively to the email ("success") is unlikely to be the same for each student since - not everyone is a fan of action thrillers, or - not everyone reads email.</td><td>Probability of picking a girl ("success") is the same for each of the 5 weekdays since all names are in the bag.</td></tr><tr><td>OR 2.</td><td>Responses may not be independent since decision of a student is likely to influence or be influenced by other students in his/her clique.</td><td>The trials of picking a student on each of the five day are independent since the names are drawn from a bag at random.</td></tr></table>		Behaviour of X	Behaviour of Y	1.	Probability of responding positively to the email ("success") is unlikely to be the same for each student since - not everyone is a fan of action thrillers, or - not everyone reads email.	Probability of picking a girl ("success") is the same for each of the 5 weekdays since all names are in the bag.	OR 2.	Responses may not be independent since decision of a student is likely to influence or be influenced by other students in his/her clique.	The trials of picking a student on each of the five day are independent since the names are drawn from a bag at random.
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11 (iii)	$Y \sim B(5, \frac{12}{21}) \text{ i.e. } Y \sim B(5, \frac{4}{7})$ $P(Y \geq 3) = 1 - P(Y \leq 2)$ $= 1 - 0.36788 \dots = 0.63212 \dots = 0.632 \text{ (3 s.f.)}$									
11 (iv)	Let W be the number of weeks in which a girl is on duty more often than a boy, out of 20. $W \sim B(20, 0.63212)$ Since $n=20$ is sufficiently large such that $np = 12.642 > 5$ and $nq = 7.3576 > 5$, $W \sim N(12.642, 4.6508) \text{ approximately.}$ $P(Y < 11) \xrightarrow{\text{continuity correction}} P(Y < 10.5)$ $= 0.16029 \dots = 0.160 \text{ (3 s.f.)}$									

	Solutions			
12(ai)	$P(A \cap B) = P(A B)P(B)$ $= \left(\frac{5}{16}x\right)x = \frac{5}{16}x^2$ $0.25 + x - \frac{5}{16}x^2 = \frac{7}{16}$ $\frac{5}{16}x^2 - x + \frac{3}{16} = 0$ $x = 0.2 \text{ or } x = 3 \text{ (reject since } 0 < x < 1)$			
12(aii)	$P(A \cap B) = \frac{5}{16}(0.2)^2 = 0.0125$ $P(A) \times P(B) = (0.25)(0.2) = 0.05$ Since $P(A) \times P(B) \neq P(A \cap B)$, thus A and B are not independent.			
12 (b)	Let A, B & C be the events a drawer containing: 2 gold coins; 1 gold coin and 1 silver coin; and 2 silver coins is selected, respectively. Let G & S be the events: a gold coin is selected; and a silver coin is selected, respectively			
12 (bi)	$P(A) = P(B) = P(C) = \frac{1}{3}$ $P(G) = P(S) = \frac{3}{6} = \frac{1}{2} \quad \text{or} \quad P(G) = \frac{1}{3} + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right) = \frac{1}{2}$			
12 (bii)	$P(\text{Drawer containing 2 gold coins is selected given that the coin selected is gold})$ $= P(A G)$ $= \frac{P(A \cap G)}{P(G)}$ $= \frac{1/3}{1/2}$ $= \frac{2}{3}$			
12 (biii)	Coins in the drawer after the older son picks a gold coin from the top drawer: <table><tr><td>Gold</td></tr><tr><td>Gold, Silver</td></tr><tr><td>Silver, Silver</td></tr></table> Required probability $= \left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)\left(\frac{1}{2}\right) = \frac{1}{2}$	Gold	Gold, Silver	Silver, Silver
Gold				
Gold, Silver				
Silver, Silver				

	Solutions
12 (biv)	P(both sons will pick a gold coin each) = P(elder son picks gold from top & younger picks gold) + P(elder picks gold from middle drawer & younger picks gold) $= \frac{1}{3} \left(\frac{1}{2} \right) + \left(\frac{1}{3} \right) \left(\frac{1}{2} \right) \left(\frac{1}{3} \right)$ $= \frac{2}{9}$
13(i)	Let A be the random variable that denotes the score of a prospective student for Kingdom University. Let B be the random variable that denotes the score of a prospective student for Island University. Given $A \sim N(500, 100^2)$ and $B \sim N(480, 60^2)$. $P(A \geq a) \leq 0.3$  <u>Alternative</u> $P(A \leq a) \geq 0.7$ $a \geq 552.44...$ The minimum score a randomly chosen student must achieve is 553 (3 s.f.).
13(ii)	$P(420 < B < 540) = 0.68268...$ $= 0.683$ (3sf) Island University has admitted 68.3% of its prospective students.
13(iii)	$\bar{A} \sim N\left(500, \frac{100^2}{5}\right)$ and $\bar{B} \sim N\left(480, \frac{60^2}{5}\right)$ $\bar{A} - \bar{B} \sim N\left(500 - 480, \frac{100^2 + 60^2}{5}\right)$ $\bar{A} - \bar{B} \sim N(20, 2720)$ <u>Method 1</u> Required Probability $= P(\bar{A} - \bar{B} < -50) + P(\bar{A} - \bar{B} > 50)$ $= 0.089767... + 0.28256...$ $= 0.37232...$ $= 0.372$ (3s.f)

	Solutions
	<u>Method 2</u> Required Probability $= P(\bar{A} - \bar{B} < -50) + P(\bar{A} - \bar{B} > 50)$ $= 1 - P(-50 < \bar{A} - \bar{B} < 50)$ $= 1 - 0.62766...$ $= 0.37234...$ $= 0.372 \text{ (3s.f.)}$
13(iv)	$B_1 + B_2 + \dots + B_5 \sim N(5 \times 480, 5 \times 60^2)$ $B_1 + B_2 + \dots + B_5 \sim N(2400, 18000)$ and $5A \sim N(5 \times 500, 5^2 \times 100^2)$ $5A \sim N(2500, 500^2)$ Let $L = B_1 + B_2 + \dots + B_5 - 5A$ $L \sim N(2400 - 2500, 18000 + 500^2)$ $L \sim N(-100, 268000)$ $P(L > 0) = 0.42341...$ $= 0.423 \text{ (3sf)}$
13(v)	Required Probability $= [P(A \geq 500)]^5 [P(B \geq 500)]^5$ $= (0.5)^5 (0.36944...) ^5$ $= 2.1506... \times 10^{-4}$ $= 2.15 \times 10^{-4} \text{ (3sf)}$ This probability found has considered <u>fewer cases</u> than the probability where the total score of these 10 prospective students is at least 5000.